B201 Computer Science Lab

Portfolio Project Report

Individual Submission - Primary Task

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BSc Computer Science

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Expected Graduation: 2028

Portfolio Website:

https://ryan-basheer-portfolio.vercel.app/

GitHub Repository:

https://github.com/ryanbasheerde7-commits/my-portfolio/

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Introduction

When I first received this assignment, I spent some time thinking about what a portfolio website really means for someone who is still in their first year of a Computer Science degree. I looked at a lot of examples online, mostly from people who had already graduated or had years of work behind them, and I quickly realised that trying to copy that kind of portfolio would not make sense for where I actually am right now.

So I decided to keep things honest and straightforward. My portfolio, available at ryan-basheer-portfolio.vercel.app, is a reflection of my current situation as a student. It shows my background, the skills I have built so far, and the two academic projects I have completed. The source code is hosted on GitHub at github.com/ryanbasheerde7-commits/my-portfolio.

The website was built using HTML5, CSS3, JavaScript and Tailwind CSS, and it is hosted on Vercel. In this report, I will walk through how the site is structured, what tools I used, the reasoning behind my design choices, and an honest reflection on what works well and what still needs improvement.

Website Structure

The portfolio is built as a single-page website. All the content is placed inside one `index.html` file, and visitors simply scroll down to read through each section rather than navigating to different pages. I chose this format for reasons I explain in the design section, but the important thing here is understanding what each part of the page contains.

Starting from the top, the first thing a visitor sees is the **Navigation Bar**. It is fixed to the top of the screen and stays visible while scrolling. It contains anchor links that jump to each section of the page, which makes it easy to move around without losing your place.

Below that is the **Hero Section**. This is the main landing area and includes my name, a short title describing me as a Computer Science student, a brief tagline, and a profile photo. The goal of this section is to give a quick and clear first impression before the visitor reads anything else.

The **About Me** section follows. This is a short paragraph written in a personal tone. It explains who I am, where I study, and why I am interested in Computer Science. I wanted this to feel like something I actually wrote rather than a copied template, so I kept it simple and direct.

After that comes the **Education** section, which lists my BSc programme at GISMA University of Applied Sciences along with the modules I have studied so far. These include Python Programming, Data Structures and Algorithms, Databases and Big Data, Mathematical Foundations, Academic Research Writing, and Project Management.

The **Technical Skills** section presents my skills in a visual grid layout, divided into three categories. The first category covers programming languages, which currently includes Python and SQL. The second covers Computer Science concepts such as Object-Oriented Programming, data structures, algorithms, Big-O notation, and database fundamentals.

The third covers tools I use regularly, including Visual Studio Code, Git, and Microsoft Office.

Next is the **Academic Projects** section. This shows two project cards, one for a Python-based Student Record Management System and one for a Relational Database Design Project using SQL. Each card has a short description of what the project involved.

The **Contact** section comes near the bottom and includes my location, email address, GitHub profile, and LinkedIn profile. It gives visitors a clear way to get in touch without too much effort.

Finally, there is a **Floating Social Island** pinned to the bottom of the screen. This is a small bar containing quick links to my social profiles. It stays visible at all times no matter how far down the page the visitor has scrolled.

GitHub Repository Structure

The repository is kept simple and easy to follow. There are no complex build tools or configuration files involved. The structure is as follows:

```
portfolio/
|
|-- index.html
|-- style.css
|-- script.js
|-- README.md
|
|-- images/
|    |-- ryan-profile.png
|    |-- title-image.png
|
|-- documents/
        |-- Ryan-CV.pdf
```

The `index.html` file contains all the HTML for every section of the page. The `style.css` file holds custom CSS that handles animations and specific styling that Tailwind does not cover. The `script.js` file contains the JavaScript for smooth scrolling and small interactive effects. The `images` folder stores the photos used on the site, and the `documents` folder contains my CV as a downloadable PDF file, which is linked from the portfolio.

The `README.md` file documents the project clearly. It includes the live website link, a list of features, the technologies used, and the folder structure. I believe that good documentation is part of the project itself and not something to rush through at the end.

Design Decisions

Single-Page Layout

I went with a single-page design because it felt like the most practical option for a portfolio of this size. At this stage of my degree, I have two projects to show and one page of background information. Splitting that across multiple pages would have created a lot of empty space and made the site feel incomplete. Having everything on one scrollable page keeps the experience smooth and makes it easier for someone to get a full picture of my profile without clicking around too much.

Dark Theme with Blue Accents

The colour scheme came together fairly early in the process. I spend most of my time working in Visual Studio Code, which uses a dark background by default, and most of the development tools and platforms I use follow a similar pattern. A dark background with blue accent colours felt like a natural fit for a Computer Science student's portfolio. It also happens to be easier on the eyes during long reading sessions compared to a bright white background, which I found useful when testing the site.

Tailwind CSS

I had not worked with Tailwind CSS before this project. I chose to use it after reading about how utility-first frameworks allow you to style elements directly in the HTML without writing separate CSS classes for everything. That approach saved a lot of time and kept the styling consistent across sections. Tailwind's responsive prefixes such as `sm:`, `md:`, and `lg:` also made it much easier to handle different screen sizes compared to writing media queries from scratch. For anything Tailwind could not handle on its own, such as custom animations, I added the styles manually in `style.css`.

Responsive Design

Making the site work properly on mobile was something I treated as a requirement from the beginning rather than something to fix later. A portfolio that breaks on a phone screen does not make a good impression. I checked the layout on different screen sizes throughout development and made adjustments as I went. The navigation bar collapses on smaller screens, the project cards stack vertically, and the floating social island adjusts in size so it does not block content.

Floating Social Island

I added the floating social bar after noticing something while browsing other long pages. When you are reading content halfway down the page and want to check a link or find contact details, you either have to scroll back to the top or keep scrolling to the bottom. A persistent bar at the bottom of the screen removes that problem. The social links are always visible and accessible without interrupting the reading experience. It is a small feature but one I think adds real value to the usability of the site.

Deployment with Vercel

I chose Vercel for hosting because of how well it integrates with GitHub. After connecting the repository, any change I push automatically goes live within a few seconds. There is no manual upload process or server configuration to deal with. Vercel is also free for static websites and provides a clean URL, which was important for sharing the portfolio professionally.

Technologies and Tools

Technology / Tool	Role in the Project
HTML5	Page structure and semantic markup for all sections
CSS3	Custom animations, transitions, and supplementary styles
JavaScript (ES6)	Smooth scrolling and UI interactivity
Tailwind CSS	Utility-first framework for responsive layout and styling
Font Awesome	Icons used in the skills section and social links
Google Fonts	Web typography for clean and readable text
Git	Version control throughout development
GitHub	Remote repository hosting and public code sharing
Vercel	Automatic deployment and free static site hosting
VS Code	Code editor used for all development work

I kept the technology stack as simple as possible. There is no JavaScript framework, no build pipeline, and no package manager involved. The site is just the files it needs to run, which keeps it fast, easy to update, and straightforward to understand.

Strengths and Weaknesses

Strengths

Consistent visual design. The dark and blue colour theme is applied throughout every section of the site without any inconsistencies. The fonts, spacing, and colour usage all follow the same pattern, which gives the portfolio a clean and finished look.

Works on mobile and desktop. The layout adjusts properly across different screen sizes. I tested this during development on both a desktop browser and a mobile device to make sure nothing broke or overflowed.

Loads quickly. Because the site does not use a JavaScript framework or pull in large external libraries, the page loads fast. There are no unnecessary files being downloaded in the background.

Simple to update. Each file in the project has a single clear purpose. Adding a new project card or changing a skill entry takes only a few minutes, and the update appears on the live site automatically through Vercel.

Contact information is always accessible. Between the dedicated Contact section and the floating social island, visitors can find my details at any point while browsing the page without needing to scroll to a specific location.

Weaknesses

Limited project content. I only have two projects to show at this point in my studies. The Projects section works, but it feels sparse compared to what it will eventually contain as I complete more of my degree.

JavaScript usage is limited. JavaScript is listed as one of the technologies used, but in practice its role on the site is limited to smooth scrolling and a few small UI effects. It does not yet demonstrate any real logic or interactivity beyond that.

No visitor tracking. There is no analytics tool connected to the portfolio, so there is no way to know how many people have visited, which parts of the page they spent time on, or where they came from.

Future Improvements

Looking ahead, there are several things I plan to add or improve as my skills develop over the course of my degree.

The portfolio already includes a working contact form with fields for name, email address, and message. It is connected to a backend service so that submissions go directly to my inbox. That said, I want to improve it further by adding basic input validation on the client side and a clearer confirmation message so visitors know their message was actually sent.

As I complete more projects during my studies, I will keep updating the Projects section. I am looking forward to adding work that demonstrates back-end development and database design, and eventually projects related to machine learning or artificial intelligence. Each new addition will include a description and a link to the relevant GitHub repository, following the same format already in place.

I am also considering adding a short blog or journal section to the site. The idea would be to write occasional posts about topics I am studying, problems I worked through, or things I found interesting during a particular module. This would give visitors a better sense of how I approach learning, not just what I have already completed.

On the technical side, I want to make better use of JavaScript. I have a few ideas, including animated skill progress bars that activate when the visitor scrolls to that section, and a filterable grid for the projects that lets you sort entries by language or topic. These would make JavaScript a more visible and meaningful part of the site rather than just running in the background.

I also plan to run the site through Google Lighthouse and the WAVE accessibility checker. Both tools would point to specific issues to fix, which is a more reliable approach than trying to spot problems manually.

Finally, I would like to add a toggle between dark and light mode. A lot of sites offer this option now and I think it is genuinely useful. It would also be a good exercise in using CSS custom properties and JavaScript together in a practical way.

Conclusion

This project turned out to be more involved than I initially expected, but it was also one of the most useful things I have worked on since starting my degree. Having to build something that actually functions in a browser, holds together visually, and represents me as a student pushed me to think through decisions more carefully than I usually do in coursework.

The portfolio is not finished in the sense that there is plenty more I want to add and improve. But it does what it needs to do right now. It is live, it works properly on different devices, the code is documented, and it honestly represents where I am at this point in my studies.

The most useful thing I took away from this project was not any particular technical skill. It was the habit of making deliberate decisions and being able to explain them. That is something I expect to carry forward into every project I work on from here.